**Deepesh**
**2116251901033**

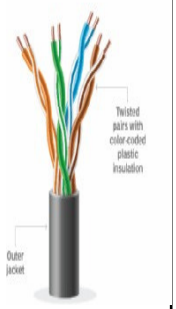
# COMPUTER NETWORKS

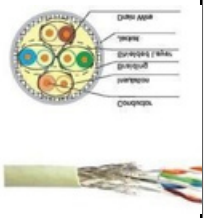
| Ex.No.2 | **Study of Network cables.** |
|---|---|
| Date: | 1. Understand different types of Network cables.<br>**2.** Make a cross-wired cable and straight through cable using a clamping/crimping tool. |

**a) Understand different types of network cable.**

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

| **Cable type** | **Category** | **Maximum Data Transmission** | **Advantages/ Disadvantages** | **Application/Use** | **Image** |
|---|---|---|---|---|---|
| UTP | Category 3 | 10 bps | **Advantages**<br>● Cheaper in cost<br>● Easy to install as they have a smaller overall diameter.<br>**Disadvantages**<br>● More prone to (EMI) Electromagnetic interference and noise | 10Base-T Ethernet |  |
| | Category 5 | Up to 100 Mbps | | Fast Ethernet, Gigabit Ethernet | |
| | Category 5e | 1Gbps | | Fast Ethernet, Gigabit Ethernet | |

| | | | | | |
|---|---|---|---|---|---|
| STP | Category6,6a | 10Gbps | **Advantages**<br>• Shielded.<br>• Faster than UTP.<br>• Less susceptible to noise and interference<br>**Disadvantages**<br>• Expensive<br>• Greater installatio n effort | Gigabit Ethernet, 10G Ethernet (55m)<br>Widely used in data centres |  |
| SSTP | Category 7 | 10Gbps | | Gigabit Ethernet, 10G Ethernet (100m) |  |

| | | | | | |
|---|---|---|---|---|---|
| Coaxial cable | RG-6<br>RG-59<br>RG-11 | 10-100Mbps | • High bandwidth<br>• Immune to interferen ce<br>• Low loss bandwidt h<br>• Versatile<br>**Disadvantages**<br>• Limited<br>• distance<br>• Cost<br>• Size is bulky<br>• | Speed of signal is 500m<br>Televisi on network<br>High speed internet connect ions |  |
| fibre optics cable | Single mode<br>Multi mode | 100Gbps | **Advantages**<br>• High speed<br>• High bandwidth<br>• High security<br>• Long distance<br>**Disadvantages**<br>• Expensive<br>• Requires | • Maximum distance of fibre optics cable is around 100meters |  |

| | | | | | |
|---|---|---|---|---|---|
| | | | skilled installers | | |

**(b) Make Your Own Ethernet Cross-Over Cable/ Straight cable**

Tools and parts needed:

- Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
- A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs.
- Optional two plug shields.

 **EthernetCable(CAT5e/CAT5)**

- CAT5e isrecommended because it supports Gigabit Ethernet (1000 Mbps).
- CAT5 alsoworks but supports lower speeds and shorter distances.

 **Crimping Tool**

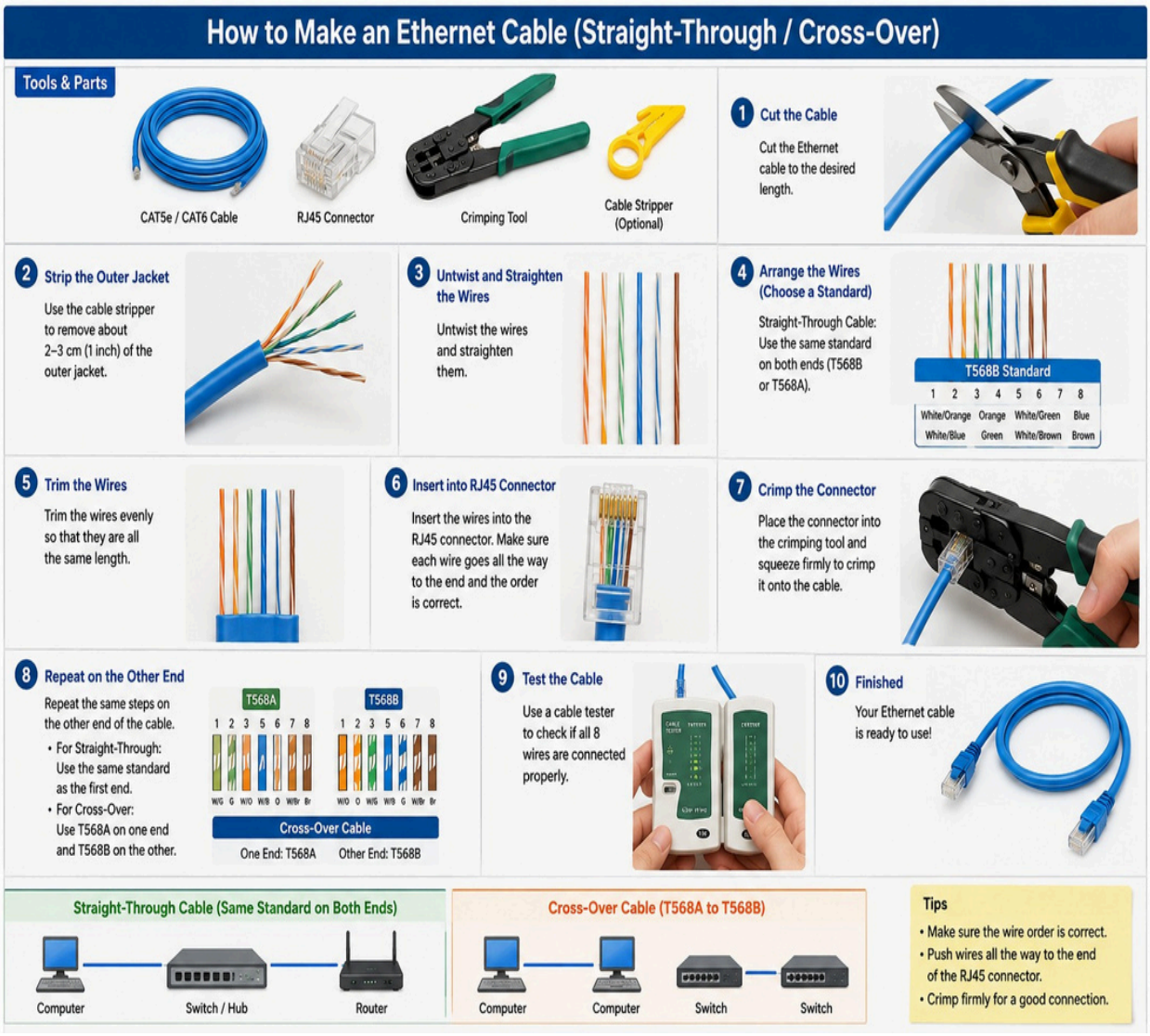
- A networkingtoolused to:
  - Stripthe outerinsulation of the cable.
  - Arrange andtrim the wires.
  - Crimp(press)the RJ45 connector onto the cable.

 **RJ45 Connectors (2)**

- Plastic connectorsfitted at both ends of the Ethernet cable.
- Used to connect thecable to computers, switches, routers, etc.

 **Plug Shields (Optional)**

- Protective covers placed over the RJ45 connectors.
- Prevent damage to the locking clip and reduce strain on the cable.




**RESULT:** Hence different types of network cables and their uses and differences has been studied.